Atoms, elements and compounds

30 questions: 15 multiple choice, 15 structured.

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1 Core Elements, compounds and mixtures [1]

Which statement correctly describes a compound?

(A) A pure substance made of only one type of atom

(B) A pure substance made of two or more elements chemically combined in a fixed ratio

(C) Two or more substances mixed together in variable proportions, without any chemical bonds between them

(D) A substance made of only one type of molecule, which can be separated into simpler substances by physical means alone

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2 Core Atomic structure and the Periodic Table [1]

In which region of an atom are protons and neutrons found?

(A) The nucleus

(B) The electron shells

(C) The outer shell only

(D) Spread evenly throughout the whole volume of the atom

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3 Core Atomic structure and the Periodic Table [1]

Which row correctly gives the relative charge and the relative mass of a neutron?

Option	Relative charge	Relative mass
A	+1	1
B	-1	1
C	0	very small (about 1/1840)
D	0	1

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4 Core Atomic structure and the Periodic Table [1]

An atom of phosphorus has a mass number of 31 and contains 16 neutrons. What is its proton (atomic) number?

A 47

B 16

C 15

D 31

5 Core Atomic structure and the Periodic Table [4]

An atom of aluminium has a proton (atomic) number of 13 and a mass (nucleon) number of 27.

(a) State the number of neutrons and the number of electrons in this atom. [2]

(b) Give the electronic configuration of this atom. [1]

(c) State the number of occupied electron shells in this atom, and identify the period of the Periodic Table in which aluminium is found. [1]

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6 Core Atomic structure and the Periodic Table [5]

(a) Describe the structure of an atom, including the location of protons, neutrons and electrons, and explain why an atom has no overall electrical charge. [3]

(b) Complete the table to give the relative charge and the relative mass of each particle.

Particle	Relative charge	Relative mass
proton		
neutron		
electron		

[2]

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7 Core Atomic structure and the Periodic Table [6]

A sulfur atom (proton number 16) gains two electrons to form a sulfide ion, S^{2-}.

(a) Deduce the number of protons and the number of electrons in the S^{2-} ion, giving a reason for your answer. [3]

(b) Give the electronic configuration of the S^{2-} ion. [1]

(c) A calcium atom (proton number 20) loses two electrons to form a Ca^{2+} ion. Give the electronic configuration of the Ca^{2+} ion, and state how many occupied electron shells it has. [2]

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8 **Core** Isotopes [1]

Which statement correctly defines isotopes?

A Atoms of the same element that have the same number of protons but different numbers of neutrons

B Atoms of different elements that have the same mass number

C Atoms of the same element that have different numbers of protons

D Molecules with the same molecular formula but different structures

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9 **Extended** Isotopes [1]

Chlorine has two common isotopes, ^{35}Cl and ^{37}Cl. Both isotopes have identical chemical properties. Which statement explains why?

A They have the same number of neutrons

B They have the same relative atomic mass

C They have the same number of protons, and therefore the same number of electrons

D They are found in different groups of the Periodic Table

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10 Core Isotopes [4]

Potassium has two isotopes, which can be written as $^{39}_{19}K$ and $^{41}_{19}K$.

(a) For the isotope $^{39}_{19}K$, state the number of protons, the number of neutrons and the number of electrons. [3]

(b) Explain, in terms of subatomic particles, why $^{39}_{19}K$ and $^{41}_{19}K$ are both isotopes of potassium. [1]

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11 **Extended** Isotopes [5]

A sample of chlorine gas contains two isotopes: 75% of the atoms are ^{35}Cl (relative isotopic mass 35) and 25% of the atoms are ^{37}Cl (relative isotopic mass 37).

(a) Calculate the relative atomic mass of this sample of chlorine. Give your answer to 3 significant figures. [3]

(b) The relative atomic mass of chlorine given in the Periodic Table is 35.5. State, with a reason, whether your answer to (a) is consistent with this value. [2]

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12 **Core** Ions and ionic bonds [1]

A magnesium atom loses two electrons when it reacts. What type of ion is formed, and what is its charge?

A An anion with charge 2-

B A cation with charge 2+

C An anion with charge 2+

D A cation with charge 2-

13 Core Ions and ionic bonds [1]

Which statement correctly defines an ionic bond?

A A shared pair of electrons between two atoms

B The attraction between a nucleus and its own electrons

C A weak attraction between neighbouring molecules

D A strong electrostatic attraction between oppositely charged ions

14 Core Ions and ionic bonds [1]

Which pair of properties is typical of an ionic compound such as sodium chloride?

A High melting point; conducts electricity when molten or in aqueous solution

B Low melting point; conducts electricity when molten or in aqueous solution

C High melting point; conducts electricity only when solid

D Low melting point; does not conduct electricity in any state

15 Extended Ions and ionic bonds [1]

Which statement best describes the structure of solid sodium chloride?

A Individual NaCl molecules held together by weak intermolecular forces

B A giant lattice of alternating Na^+ and Cl^- ions held together by strong electrostatic forces in all directions

C A small cluster of six ions arranged randomly

D A giant covalent structure with each atom bonded to four others

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16 Core Ions and ionic bonds [5]

Potassium and fluorine react together to form the ionic compound potassium fluoride, KF. A potassium atom has the electronic configuration 2,8,8,1 and a fluorine atom has the electronic configuration 2,7.

(a) Describe how a potassium atom and a fluorine atom form ions when they react together. [2]

(b) State the electronic configuration of the K^+ ion and of the F^- ion. [2]

(c) Explain why potassium fluoride is described as having an ionic bond. [1]

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17 Core Ions and ionic bonds [4]

(a) State two physical properties of solid sodium chloride that are typical of ionic compounds. [2]

(b) State whether solid sodium chloride conducts electricity, and state whether molten sodium chloride conducts electricity. [2]

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18 **Extended** Ions and ionic bonds [6]

Magnesium (electronic configuration 2,8,2) reacts with oxygen (electronic configuration 2,6) to form magnesium oxide, MgO.

(a) Describe how a magnesium atom and an oxygen atom form ions in this reaction. [2]

(b) Give the electronic configuration of the Mg^{2+} ion and of the O^{2-} ion. [2]

(c) Explain, in terms of electron transfer, why the ratio of magnesium ions to oxide ions in magnesium oxide is 1:1. [2]

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19 **Extended** Ions and ionic bonds [6]

Calcium oxide, CaO, is an ionic compound.

(a) Describe the structure of calcium oxide as a giant ionic lattice. [2]

(b) Explain, in terms of structure and bonding, why calcium oxide has a high melting point. [2]

(c) Explain, in terms of structure and bonding, why calcium oxide conducts electricity when molten but not when solid. [2]

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20 **Core** Simple molecules and covalent bonds [1]

What is a covalent bond?

A An electrostatic attraction between oppositely charged ions

B A pair of electrons transferred completely from one atom to another

C A pair of electrons shared between two atoms

D An attraction between a metal ion and a sea of delocalised electrons

21 Core Simple molecules and covalent bonds [1]

Which pair of properties is typical of a simple molecular compound such as water?

A High melting point; good electrical conductivity

B High melting point; poor electrical conductivity

C Low melting point; good electrical conductivity

D Low melting point; poor electrical conductivity

22 Core Simple molecules and covalent bonds [4]

Methane, CH_4, is a simple molecular compound.

(a) State the number of covalent bonds formed by the carbon atom in a molecule of methane, and state the number of electrons shared in each of these bonds. [2]

(b) Describe, in terms of electron arrangement, how the covalent bonds in methane give both the carbon atom and each hydrogen atom a full outer shell of electrons. [2]

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23 **Extended** Simple molecules and covalent bonds [5]

In a molecule of carbon dioxide, CO_2, the carbon atom forms a double covalent bond with each of the two oxygen atoms.

(a) Describe, in terms of shared electrons, the covalent bonding between the carbon atom and each oxygen atom in a molecule of carbon dioxide. [3]

(b) In a molecule of nitrogen, N_2, the two nitrogen atoms share electrons so that each nitrogen atom has a full outer shell. State the number of electrons shared between the two nitrogen atoms, and name the type of covalent bond formed. [2]

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24 **Extended** Simple molecules and covalent bonds [6]

Iodine, I_2, is a simple molecular substance.

(a) Explain, in terms of structure and bonding, why simple molecular substances such as iodine have low melting points and low boiling points. [3]

(b) Explain why simple molecular substances are poor conductors of electricity. [2]

(c) State one way in which the covalent bond within a molecule of iodine differs from the forces between neighbouring molecules of iodine. [1]

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25 Core Giant covalent structures [1]

In which structure is each carbon atom covalently bonded to four other carbon atoms?

A Diamond

B Graphite

C Carbon dioxide

D Methane

26 Core Giant covalent structures [1]

Which property of graphite makes it useful as an electrode in electrolysis?

A It is very hard and resists scratching

B It conducts electricity and has a high melting point

C It is transparent and colourless

D It is soft and dissolves easily in water

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27 Core Giant covalent structures [6]

(a) Describe the structure and bonding in diamond. [2]

(b) Describe the structure and bonding in graphite, and explain how this structure allows graphite to be used as a lubricant. [2]

(c) State one other use of graphite and one use of diamond, and for each one link the use to a property of its structure. [2]

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28 **Extended** Giant covalent structures [5]

Silicon(IV) oxide, SiO_2, has a giant covalent structure similar to diamond.

(a) Describe the structure of silicon(IV) oxide. [2]

(b) Explain why silicon(IV) oxide, like diamond, has a very high melting point and does not conduct electricity. [3]

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29 **Extended** Metallic bonding [1]

Which description best fits the structure of a metal such as copper?

A A lattice of alternating positive and negative ions

B Individual atoms held together by weak intermolecular forces

C A giant lattice of positive ions surrounded by a sea of delocalised electrons

D A lattice of neutral atoms joined by covalent bonds

30 **Extended** Metallic bonding [6]

(a) Describe the structure of metallic bonding in a metal such as copper. [2]

(b) Explain, in terms of structure and bonding, why metals are good conductors of electricity. [2]

(c) Explain, in terms of structure and bonding, why metals can be bent and shaped without breaking. [2]

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